Cambridge Assessment International Education

# Cambridge IGCSE™

**BIOLOGY** **0610/11**

Paper 1 Multiple Choice (Core) **October/November 2020**

**45 minutes**

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

**INSTRUCTIONS**

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

**INFORMATION**

- The total mark for this paper is 40.
- Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
- Any rough working should be done on this question paper.

This document has **16** pages. Blank pages are indicated.

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**[Turn over**

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**1** Hedgehogs are mammals. Touching a hedgehog causes it to roll into a ball to protect itself.

Which characteristics is it displaying?

**A** excretion and movement

**B** growth and sensitivity

**C** movement and growth

**D** movement and sensitivity

**2** According to the binomial system, how should a human be named?

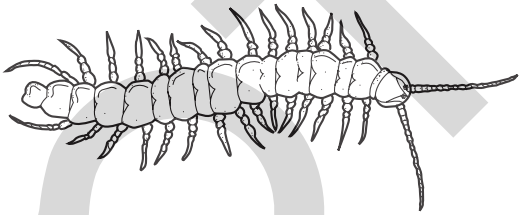
**A** *Homo Sapiens*

**B** *Homo sapiens*

**C** *homo Sapiens*

**D** *homo sapiens*

**3** The diagram shows an arthropod.



Which group does this arthropod belong to?

**A** arachnid

**B** crustacean

**C** insect

**D** myriapod

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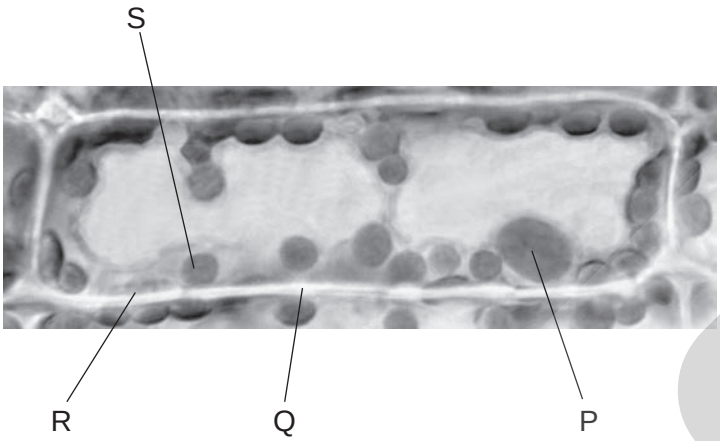
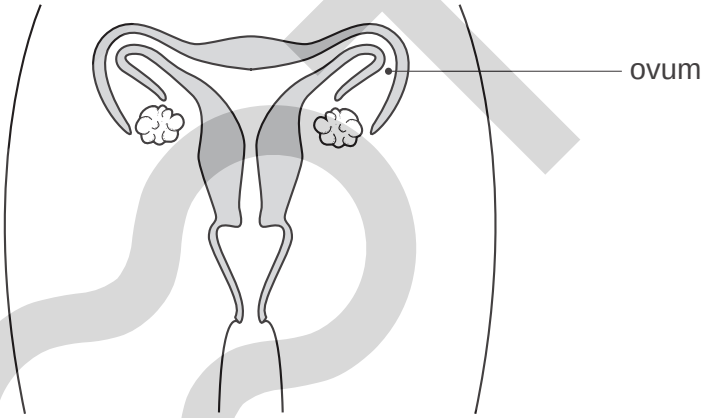
**4** The photomicrograph shows a cell from a type of aquatic plant.




Which parts labelled on the photomicrograph indicate that this is a plant cell?

**A** P and R **B** P and S **C** Q and R **D** Q and S

**5** The diagram shows the female reproductive system.




At which level of organisation are the ovum and the ovary?

| | ovum | ovary |
|---|---|---|
| **A** | cell | organ |
| **B** | cell | tissue |
| **C** | organ | tissue |
| **D** | tissue | organ |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**6** The cell shown in the diagram has been magnified 3000 times. The diagram is 21 mm wide.

What is the actual diameter of the cell?

**A** 21 mm

**B** $\frac{21\,\text{mm}}{3000}$

**C** 21 mm × 3000

**D** $\frac{3000}{21\,\text{mm}}$

**7** What are features of osmosis?

| | diffusion is involved | requires cell walls | requires a partially permeable membrane |
|---|---|---|---|
| **A** | ✓ | ✗ | ✓ |
| **B** | ✓ | ✗ | ✗ |
| **C** | ✗ | ✓ | ✓ |
| **D** | ✗ | ✓ | ✗ |

key
✓ = yes
✗ = no

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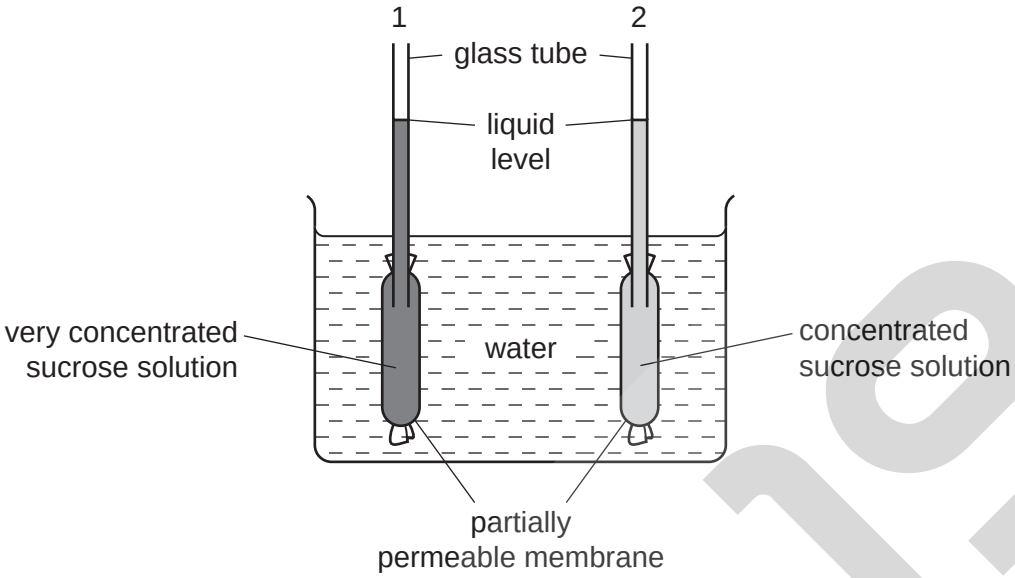
**8** The diagram shows apparatus which can be used to demonstrate osmosis.




After one hour, what would happen to the liquid levels in the glass tubes?

| | liquid level in tube 1 | liquid level in tube 2 |
|---|---|---|
| **A** | falls | falls |
| **B** | falls | rises |
| **C** | rises | falls |
| **D** | rises | rises |

**9** Which substances are made by linking together glucose molecules only?

**A** cellulose, glycogen and starch

**B** fats, cellulose and proteins

**C** proteins, oils and glycogen

**D** starch, fats and oils

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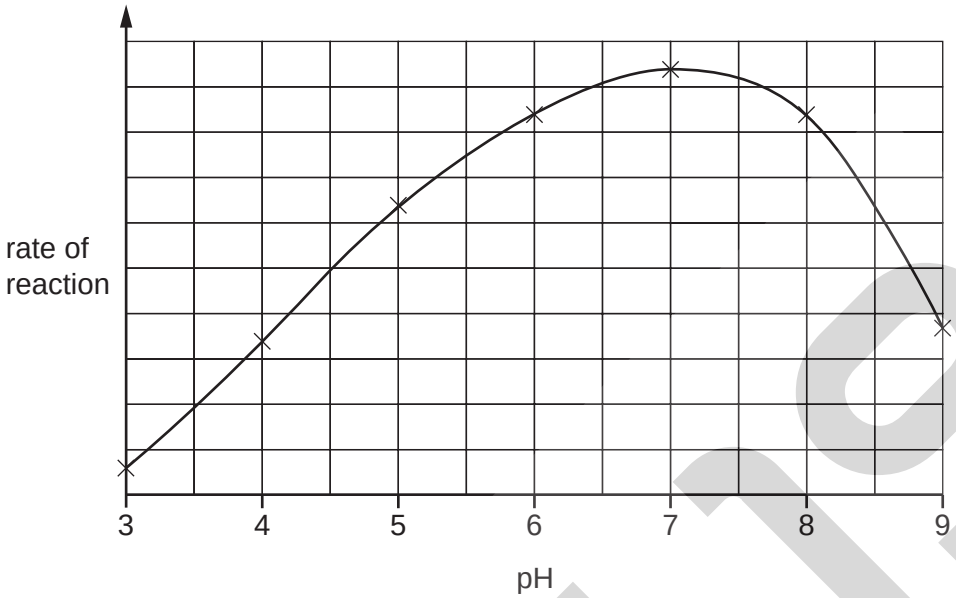
**10** The graph shows the effect of pH on the rate of reaction of an enzyme.




What does the graph show?

**A** The enzyme is destroyed at pH 9.

**B** The enzyme works best at pH 6.

**C** The rate of reaction halves as the pH changes from pH 5 to pH 7.

**D** The rate of reaction is the same at pH 5 and pH 8.5.

**11** What kind of molecule is an enzyme?

**A** fat

**B** glucose

**C** protein

**D** starch

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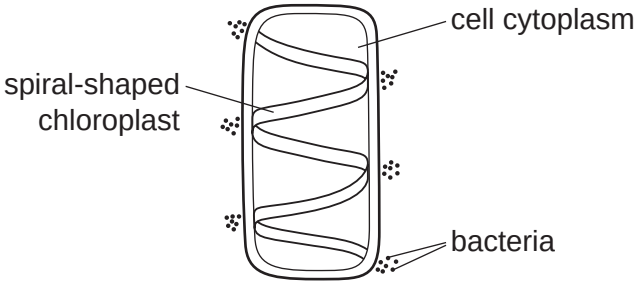
**12** The diagram shows a cell with groups of bacteria around its edge.




The bacteria move to areas of high oxygen concentration.

Which process in the cell causes the bacteria to form these groups?

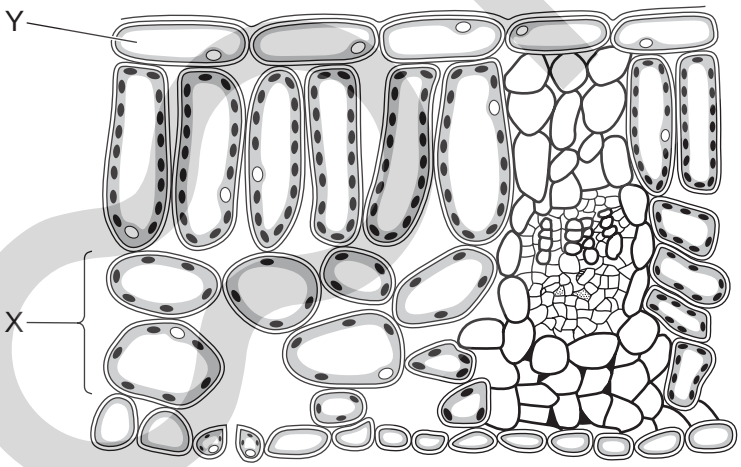
**A** digestion

**B** photosynthesis

**C** reproduction

**D** respiration

**13** The diagram shows a section through a leaf.




Which row correctly identifies tissues X and Y?

| | X | Y |
|---|---|---|
| **A** | palisade mesophyll | cuticle |
| **B** | palisade mesophyll | upper epidermis |
| **C** | spongy mesophyll | cuticle |
| **D** | spongy mesophyll | upper epidermis |

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**14** Which stage of nutrition takes place when food molecules become part of a body cell?

**A** absorption

**B** assimilation

**C** digestion

**D** ingestion

**15** What is the definition of chemical digestion?

**A** Large insoluble molecules are changed into smaller soluble molecules.

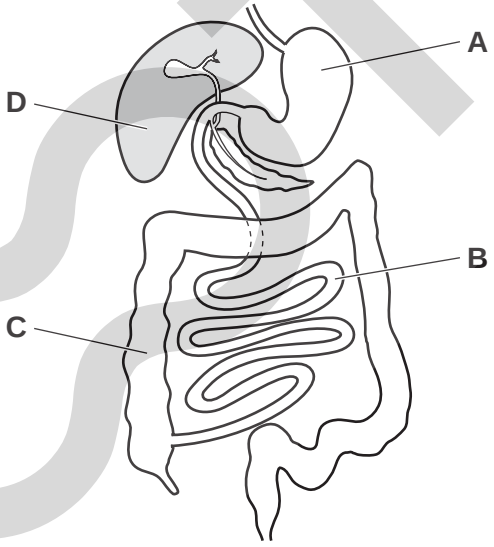
**B** Large soluble molecules are changed into smaller soluble molecules.

**C** Small insoluble molecules and ions are passed through the wall of the small intestine.

**D** Small soluble molecules and ions are passed through the wall of the small intestine.

**16** The diagram shows the human alimentary canal.

Where is most digested food absorbed into the blood?




**17** In which order does water pass through these structures in a plant?

**A** mesophyll → root hair → xylem

**B** mesophyll → xylem → root hair

**C** root hair → mesophyll → xylem

**D** root hair → xylem → mesophyll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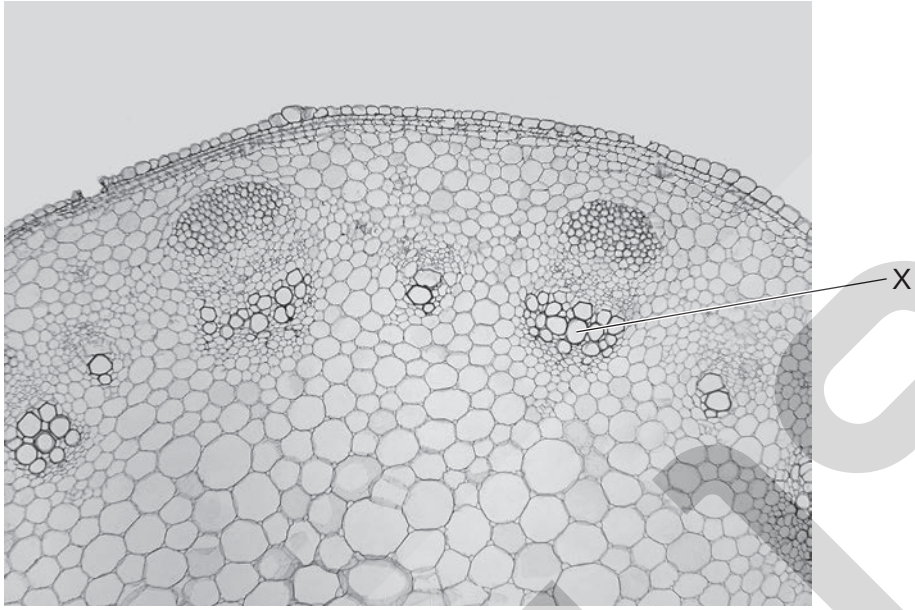
**18** The photomicrograph shows a cross-section through a plant stem.




What is the name and function of the tissue labelled X?

| | tissue | function |
|---|---|---|
| **A** | phloem | transports sugars |
| **B** | phloem | transports water and minerals |
| **C** | xylem | transports sugars |
| **D** | xylem | transports water and minerals |

**19** The table shows some features of blood vessels.

Which row shows the features of a vein?

| | direction of blood flow | size of the central hole (lumen) | thickness of wall |
|---|---|---|---|
| **A** | away from the heart | large | thick |
| **B** | away from the heart | small | thick |
| **C** | towards heart | small | thin |
| **D** | towards heart | large | thin |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**20** Which component of the blood releases antibodies in response to an infection by bacteria?

**A** blood plasma

**B** platelets

**C** red blood cells

**D** white blood cells

**21** Which is an example of a transmissible disease?

**A** coronary heart disease caused by eating a high fat diet

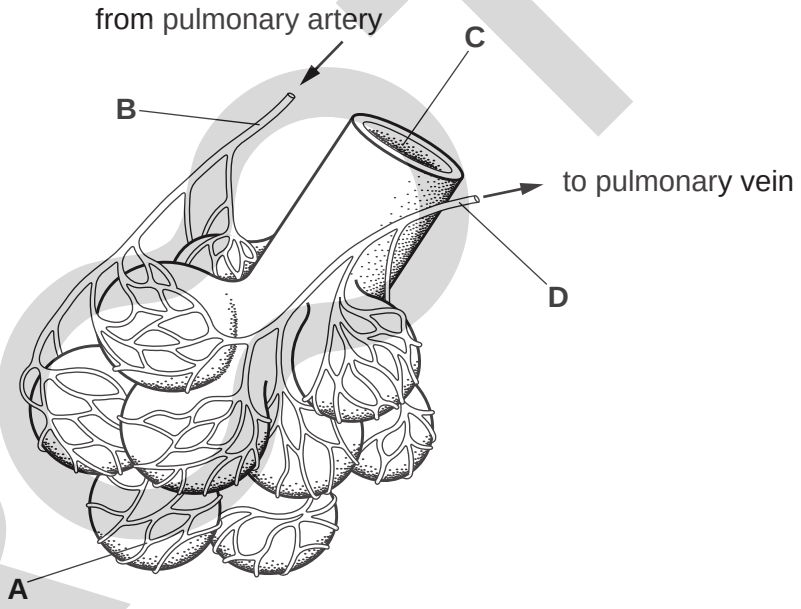
**B** AIDS

**C** liver disease caused by excessive alcohol consumption

**D** scurvy

**22** The diagram shows some of the structures in a human lung.

Where is the carbon dioxide concentration highest?




**23** The list shows some processes that happen in the human body.

1 water enters cells by osmosis

2 muscles contract

3 impulses travel along neurones

4 oxygen diffuses into cells

Which of these require energy released by respiration?

**A** 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2, 3 and 4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**24** The amount of urea in the blood increases as it passes through organ X.

What is organ X?

**A** heart

**B** kidney

**C** liver

**D** pancreas

**25** Which is the correct sequence of structures through which a nerve impulse passes in a reflex arc?

**A** effector → motor neurone → relay neurone → sensory neurone → receptor

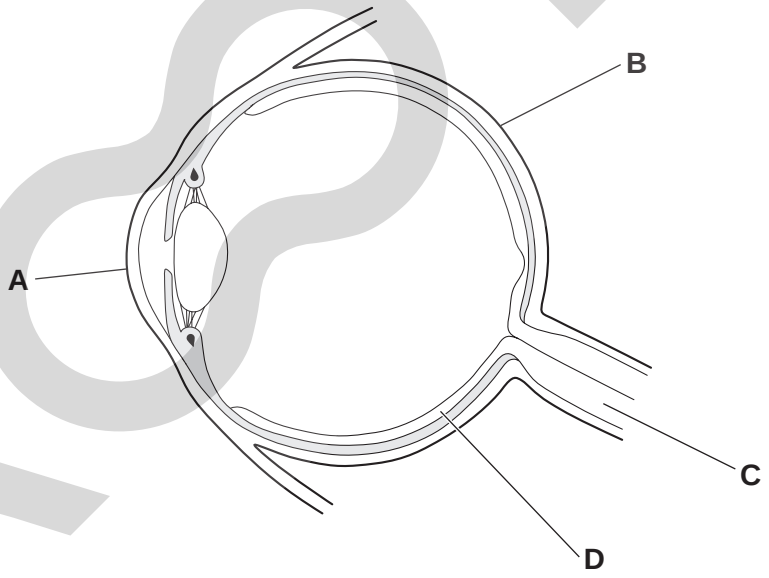
**B** effector → sensory neurone → relay neurone → motor neurone → receptor

**C** receptor → motor neurone → relay neurone → sensory neurone → effector

**D** receptor → sensory neurone → relay neurone → motor neurone → effector

**26** The diagram shows an eye.

Which labelled part contains cells that are sensitive to light?




**27** Which hormone stimulates the development of male gametes?

**A** adrenaline

**B** insulin

**C** oestrogen

**D** testosterone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**28** A drug is defined as a substance taken into the body that

**A** causes addiction.

**B** cures disease.

**C** kills bacteria.

**D** modifies or affects chemical reactions.

**29** What is an example of asexual reproduction?

**A** a single bacterium dividing to produce two genetically identical bacteria

**B** a mammal giving birth to four offspring

**C** a female bird laying six eggs which will hatch into the same species

**D** a tomato plant producing fruits which contain many seeds

**30** What is the sequence of events when a woman gives birth?

| | 1st | 2nd | 3rd |
|---|---|---|---|
| **A** | delivery of afterbirth | cervix dilates | passage of fetus through vagina |
| **B** | cervix dilates | passage of fetus through vagina | umbilical cord cut |
| **C** | passage of fetus through vagina | amniotic sac breaks | cervix dilates |
| **D** | umbilical cord cut | passage of fetus through vagina | delivery of afterbirth |

**31** Some types of birth control are listed.

1 condom

2 diaphragm

3 IUD

4 vasectomy

Which types are barrier methods?

**A** 1, 2 and 4 **B** 1 and 2 **C** 1 only **D** 3 and 4

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**32** In humans, sex determination is controlled by the X and Y chromosomes.

A man and a woman have three children, two boys and a girl.

What is the probability that their next child will be a girl?

**A** 25% **B** 50% **C** 75% **D** 100%

**33** Mitosis is a type of cell division.

Which row about mitosis is correct?

| | cells produced | functions |
|---|---|---|
| **A** | genetically different | growth and repair of tissues |
| **B** | genetically different | production of gametes |
| **C** | genetically identical | growth and repair of tissues |
| **D** | genetically identical | production of gametes |

**34** Which statement is best used to describe differences in height in humans?

**A** continuous variation resulting in a limited number of phenotypes

**B** continuous variation resulting in a range of phenotypes

**C** discontinuous variation resulting in a limited number of phenotypes

**D** discontinuous variation resulting in a range of phenotypes

**35** A student wrote some statements describing the stages involved during selective breeding of plants.

1 Farmers select individual plants with desirable features.

2 Farmers cross these individuals to produce the next generation.

3 The farmers then select the offspring with the desirable features.

Which statements are correct?

**A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

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**36** The diagram shows a food chain.

clover plant → rabbit → fox

Which terms can be used to describe the fox in this food chain?

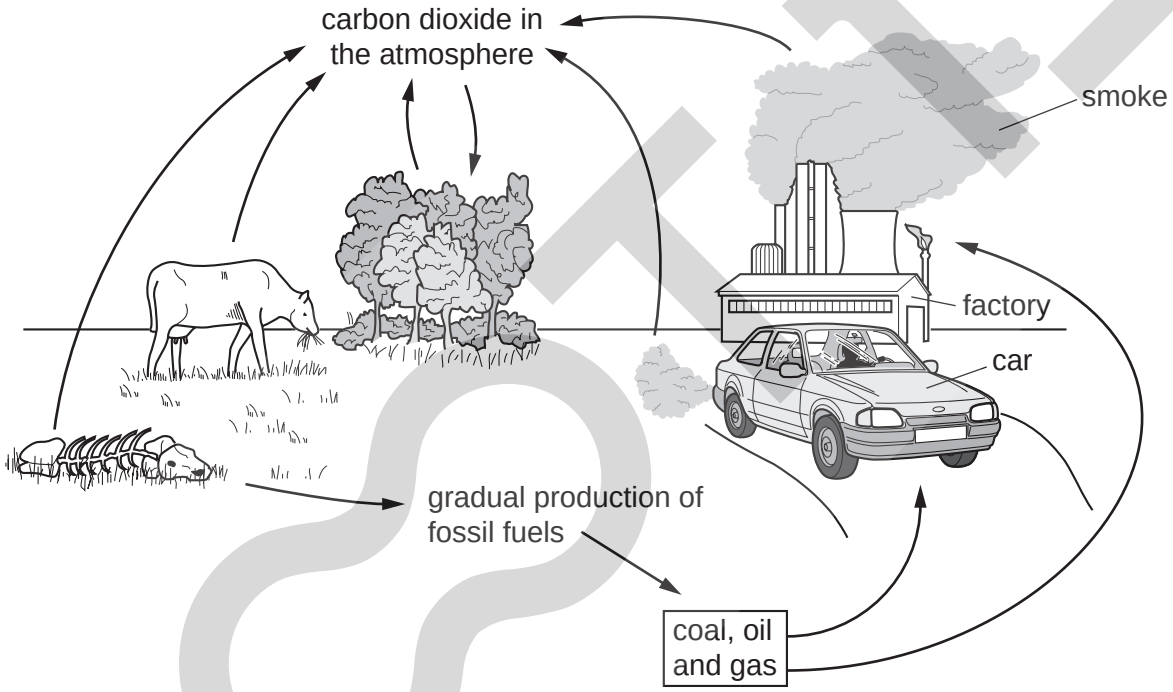
**A** carnivore and primary consumer

**B** carnivore and secondary consumer

**C** carnivore and tertiary consumer

**D** carnivore only

**37** The diagram represents the carbon cycle.




Which process produces the carbon dioxide from the factory and the car?

**A** combustion

**B** condensation

**C** decomposition

**D** precipitation

**38** In the water cycle, which stage is **only** carried out by plants?

**A** condensation

**B** evaporation

**C** precipitation

**D** transpiration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**39** Yeast carries out anaerobic respiration, making carbon dioxide and ethanol as end products.

Which end products can be used to make biofuel and bread?

| | biofuel | bread |
|---|---|---|
| **A** | carbon dioxide | carbon dioxide |
| **B** | carbon dioxide | ethanol |
| **C** | ethanol | carbon dioxide |
| **D** | ethanol | ethanol |

**40** What is the effect of replacing tropical rainforest with plantations of oil palm trees, which are grown as crop plants?

**A** increased carbon dioxide in the atmosphere

**B** increased flooding

**C** reduced number of species

**D** reduced pollution

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